

Homefield Advantage Design Document

Program Overview:

The Homefield Advantage java file is program that accepts a valid text file and analyzes the data line by line to see whether being on the homefield really has an advantage or not.

- Using the Information from these Analysis, we will write a report stating the validity of the hypothesis that being on the homefield has an advantage over the team's ability to win or not.
- Inputs:
 - Keyboard (Scanner for System.in)
 - Files (Scanner for File)
- Outputs:
 - Prints the Result to the Console

Variables to Keep Track of:

- Total Number of Games
- Total Number of Home Wins
- Total Number of Home Games
- Total and Average Margin

English Overview:

1. Intro
2. Ask for Valid File
3. Go through File
4. Analyze and Print Results
5. Ask the user if they want to play it again.

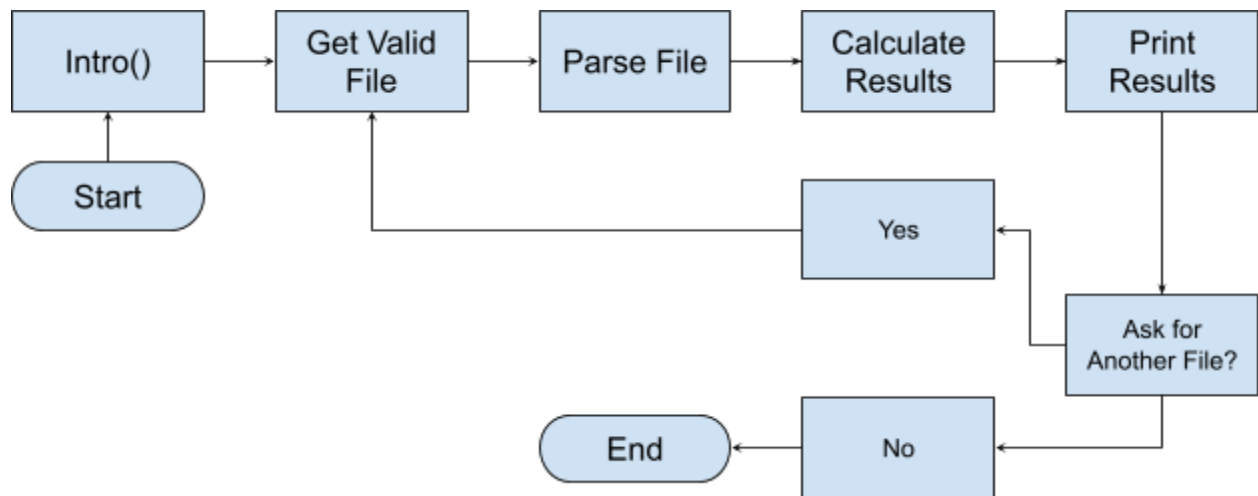
How to Parse Each Game Line

2007-11-11 @Texas 92 Missouri St 55

Steps:

1. Token Based Processing
2. Skip Date
3. Get Team Names
4. Get Team Scores
5. Determine if it is a Home Team or Not.
6. Update Stats.

Program Steps:



Example Output per File:

HOME FIELD ADVANTAGE RESULTS

Total number of games: 2,467

Number of games with a home team: 2,465

Percentage of games with a home team: 99.9%

Number of games the home team won: 1,312

Home team win percentage: 53.2%

Home team average margin: 0.16

Functions:

- main()
 - Calls intro()
 - Loops:
 - Calls getFile()
 - Calls analyzeFile()
 - Calls printResults()
 - Prompt to see if you want to have another
- getFile()
 - Prompts until a valid filename is entered.
 - Returns a Scanner for File.
- analyzeFile()
 - Reads the first line: Sport
 - Loops over each game line.

- Checks for @ for home games
 - Returns totals for stores them in local vars.
- printResults()
 - Uses printf for proper formatting
 - Prints the required lines.